# Model vs Index — CAGR Comparison

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## What period this covers

| Item | Value |
|---|---|
| Test window | 2018-11-01 to 2020-06-30 |
| Rebalance frequency | monthly |
| Rebalance dates | 19 |
| Elapsed time | 1.58 years |
| Holding period | entry to the next rebalance's entry (chaining) |
| Portfolio | equal-weighted top 20, rebalanced each period |
| CAGR formula | product of (1 + period return) over 19 periods, then raised to 1/1.58 |

## Basis

**Price return compared with price return.** NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

**Index returns are measured over the sessions the strategy actually traded** — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

## Headline: CAGR by strategy

| Strategy | Gross | Net | vs UNIVERSE | vs N50 | vs N500 | vs NMIDCAP 150 | vs NSMALLCAP 250 |
|---|---|---|---|---|---|---|---|
| equal weight universe | -10.95% | -12.81% | -1.86% | -10.16% | -8.80% | -6.72% | +0.73% |
| growth only | -16.02% | -19.94% | -8.98% | -17.28% | -15.93% | -13.85% | -6.40% |
| model 5 | 19.71% | 11.46% | +22.42% | +14.12% | +15.47% | +17.55% | +25.00% |
| model 5 gated | 19.71% | 11.46% | +22.42% | +14.12% | +15.47% | +17.55% | +25.00% |
| momentum only | 10.71% | 1.78% | +12.73% | +4.43% | +5.79% | +7.87% | +15.32% |
| quality only | -4.68% | -12.35% | -1.39% | -9.69% | -8.34% | -6.26% | +1.19% |
| random ranking | -14.60% | -34.81% | -23.86% | -32.16% | -30.80% | -28.72% | -21.27% |
| simple quality momentum | 7.89% | 0.44% | +11.39% | +3.09% | +4.45% | +6.53% | +13.98% |
| value only | 21.24% | 16.45% | +27.40% | +19.10% | +20.46% | +22.54% | +29.99% |

**vs UNIVERSE is the column that matters, and it is deliberately placed before the index columns.** It compares the strategy against an equal-weighted holding of the very same point-in-time universe it selected from, so it isolates stock selection.

**Why the index columns flatter every strategy.** The eligible universe returned -10.95% against NIFTY 500's -4.01% — a -6.95 point gap earned purely by equal-weighting a broad, small-cap-heavy universe, before any selection at all. That gap is inside every "vs index" number below. A strategy that beats NIFTY 500 by 20 points while matching the universe has selected nothing.

## Index reference

| Index | CAGR | Volatility | Sharpe | Max drawdown | Periods |
|---|---|---|---|---|---|
| NIFTY 50 | -2.65% | 27.84% | 0.05 | -32.25% | 19 |
| NIFTY 500 | -4.01% | 28.16% | 0.01 | -31.85% | 19 |
| NIFTY MIDCAP 150 | -6.09% | 31.34% | -0.03 | -32.75% | 19 |

| | | | | | |
|---|---|---|---|---|---|
| NIFTY SMALLCAP 250 | -13.54% | 35.09% | -0.22 | -42.26% | 19 |
| Eligible universe (equal weight) | -10.95% | - | - | - | 19 |

The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

## Risk-adjusted comparison

| Strategy | Basis | CAGR | Volatility | Sharpe | Max DD | Hit rate |
|---|---|---|---|---|---|---|
| equal weight universe | net | -12.81% | 36.05% | -0.19 | -43.18% | 58% |
| growth only | net | -19.94% | 43.40% | -0.29 | -54.60% | 47% |
| model 5 | net | 11.46% | 23.16% | 0.58 | -16.04% | 63% |
| model 5 gated | net | 11.46% | 23.16% | 0.58 | -16.04% | 63% |
| momentum only | net | 1.78% | 25.01% | 0.19 | -21.27% | 53% |
| quality only | net | -12.35% | 38.75% | -0.14 | -43.92% | 47% |
| random ranking | net | -34.81% | 40.12% | -0.83 | -62.67% | 42% |
| simple quality momentum | net | 0.44% | 24.50% | 0.14 | -23.92% | 42% |
| value only | net | 16.45% | 47.18% | 0.54 | -33.56% | 47% |

## Excess return against NIFTY 500

| Strategy | Mean excess/period | Tracking error | Information ratio | Periods beaten |
|---|---|---|---|---|
| equal weight universe | -0.60% | 13.95% | -0.51 | 26% |
| growth only | -1.06% | 21.97% | -0.58 | 32% |
| model 5 | +1.11% | 14.46% | 0.92 | 63% |
| model 5 gated | +1.11% | 14.46% | 0.92 | 63% |
| momentum only | +0.39% | 17.51% | 0.26 | 37% |
| quality only | -0.47% | 18.22% | -0.31 | 47% |
| random ranking | -2.80% | 18.55% | -1.81 | 37% |
| simple quality momentum | +0.26% | 17.63% | 0.18 | 58% |
| value only | +2.11% | 30.37% | 0.84 | 58% |

**Information ratio** is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

## Turnover and cost drag

| Strategy | One-way turnover | Gross CAGR | Net CAGR | Cost drag |
|---|---|---|---|---|
| equal weight universe | 0.087 | -10.95% | -12.81% | 1.86% |
| growth only | 0.163 | -16.02% | -19.94% | 3.92% |
| model 5 | 0.385 | 19.71% | 11.46% | 8.24% |
| model 5 gated | 0.385 | 19.71% | 11.46% | 8.24% |
| momentum only | 0.468 | 10.71% | 1.78% | 8.93% |
| quality only | 0.245 | -4.68% | -12.35% | 7.67% |
| random ranking | 0.950 | -14.60% | -34.81% | 20.21% |
| simple quality momentum | 0.398 | 7.89% | 0.44% | 7.45% |
| value only | 0.158 | 21.24% | 16.45% | 4.79% |

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

## Before quoting these numbers

**An index is investable and this portfolio is a simulation.** The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

**The window is what it is.** A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

**Costs are modelled, not measured.** Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

**Delisting recovery is an assumption.** Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.